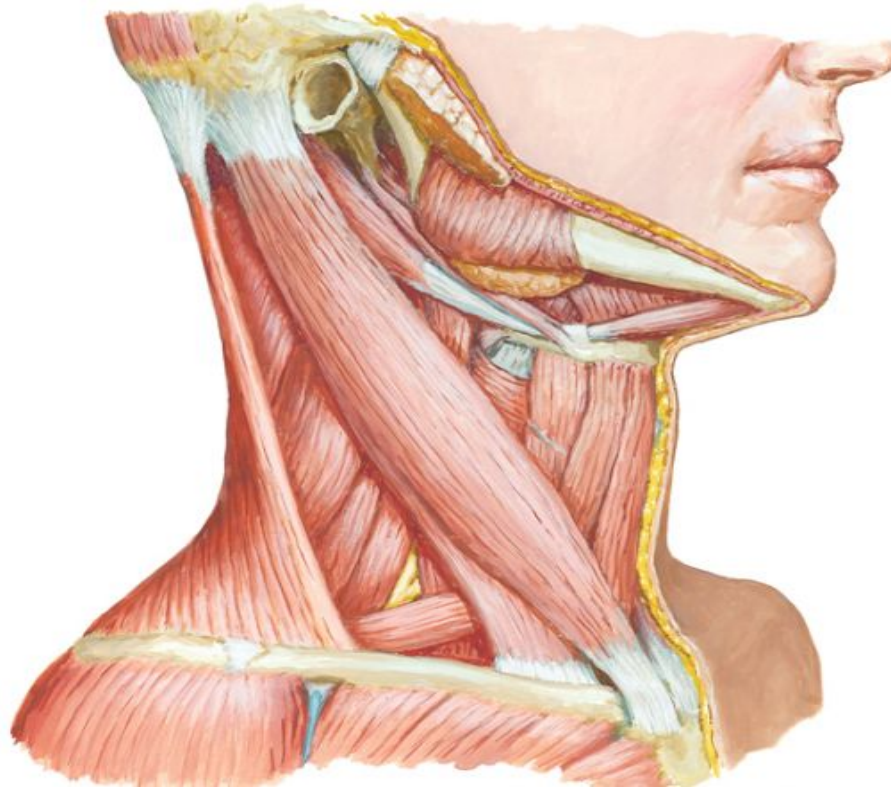
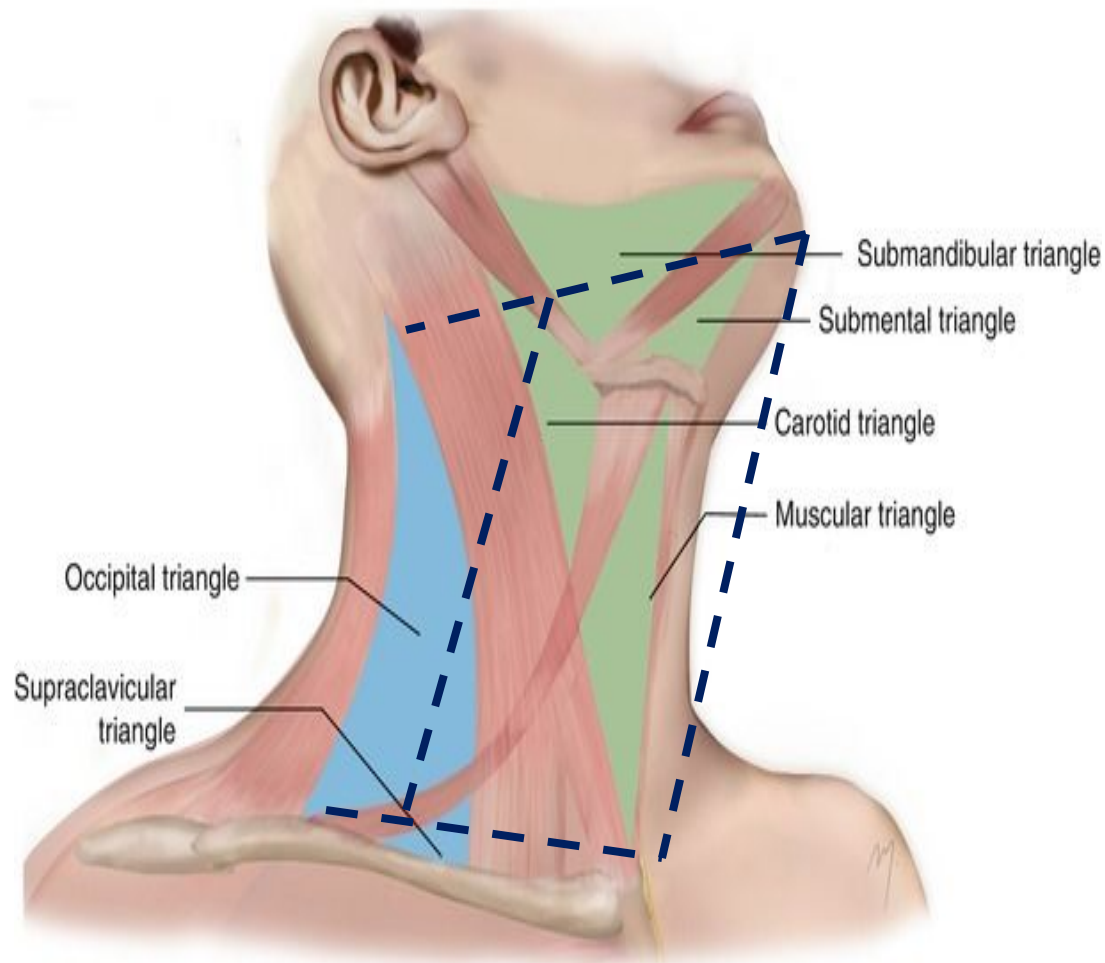


# Posterior triangle of the neck



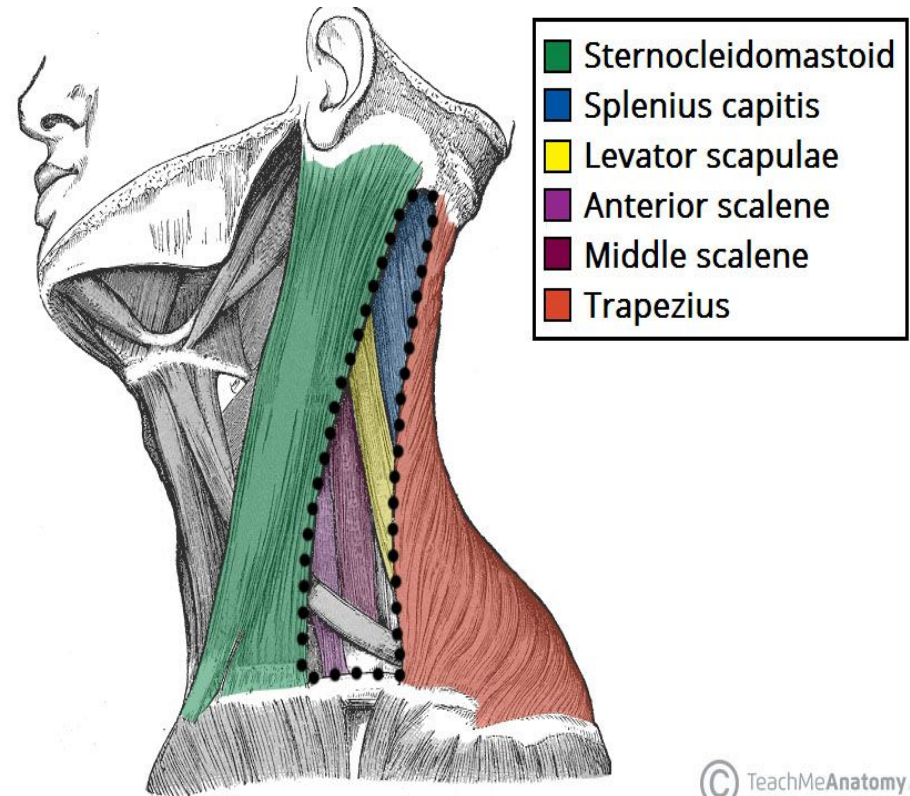
# Introduction:

- The side of the neck is almost quadrilateral in outline.
- It is bounded anteriorly by anterior midline,
- Posteriorly by anterior border of trapezius,
- Superiorly by base of mandible, a line joining the angle of mandible to the mastoid process and superior nuchal line,
- Inferiorly by clavicle.

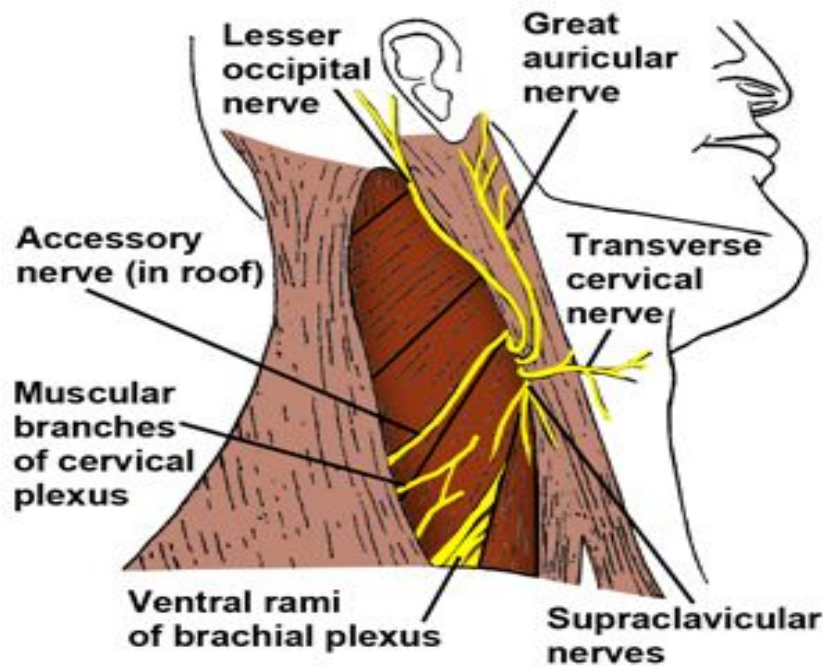


# Boundaries of Posterior Triangle:

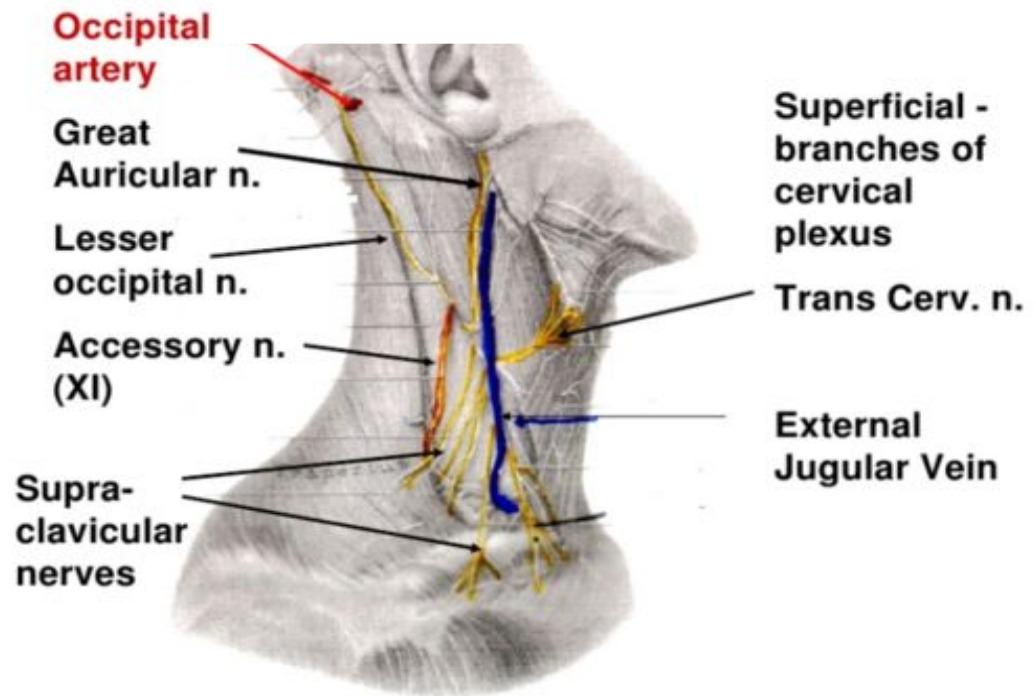
- The posterior triangle is a space on the side of the neck located behind sternocleidomastoid.
- **Boundaries:**
  - **Anterior:** posterior border of sternocleidomastoid
  - **Posterior:** anterior border of trapezius
  - **Inferior:** middle one-third of clavicle
  - **Apex:** lies at superior nuchal line where sternomastoid and trapezius meet



**Roof:** is formed by investing layer of deep cervical fascia, superficial fascia containing platysma, external jugular vein and posterior external jugular vein



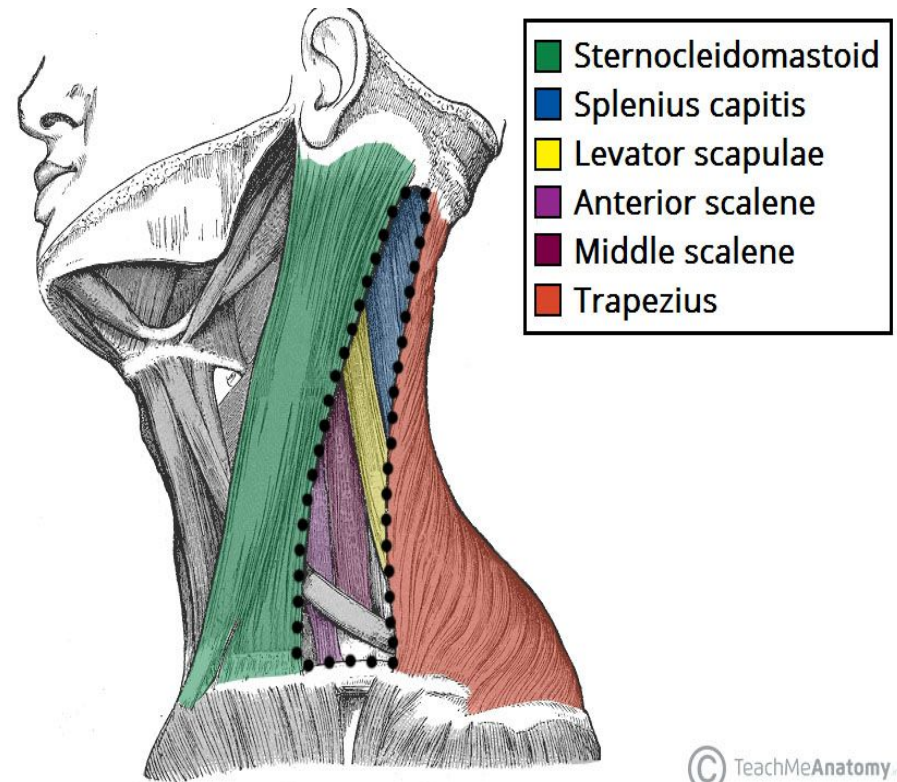
#### B. CONTENTS OF POSTERIOR TRIANGLE





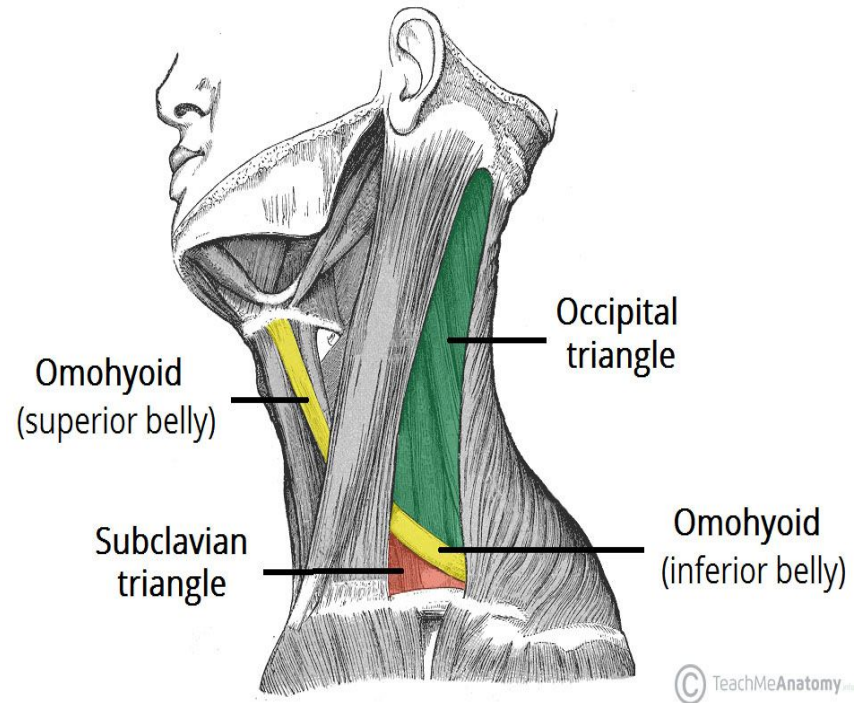
□ **Floor:** is formed by prevertebral layer of deep cervical fascia and the underlying muscles as follows:

- I. Splenius capitis
- II. Levator scapulae
- III. Scalenus medius
- IV. Scalenus anterior



# Divisions of Posterior Triangle:

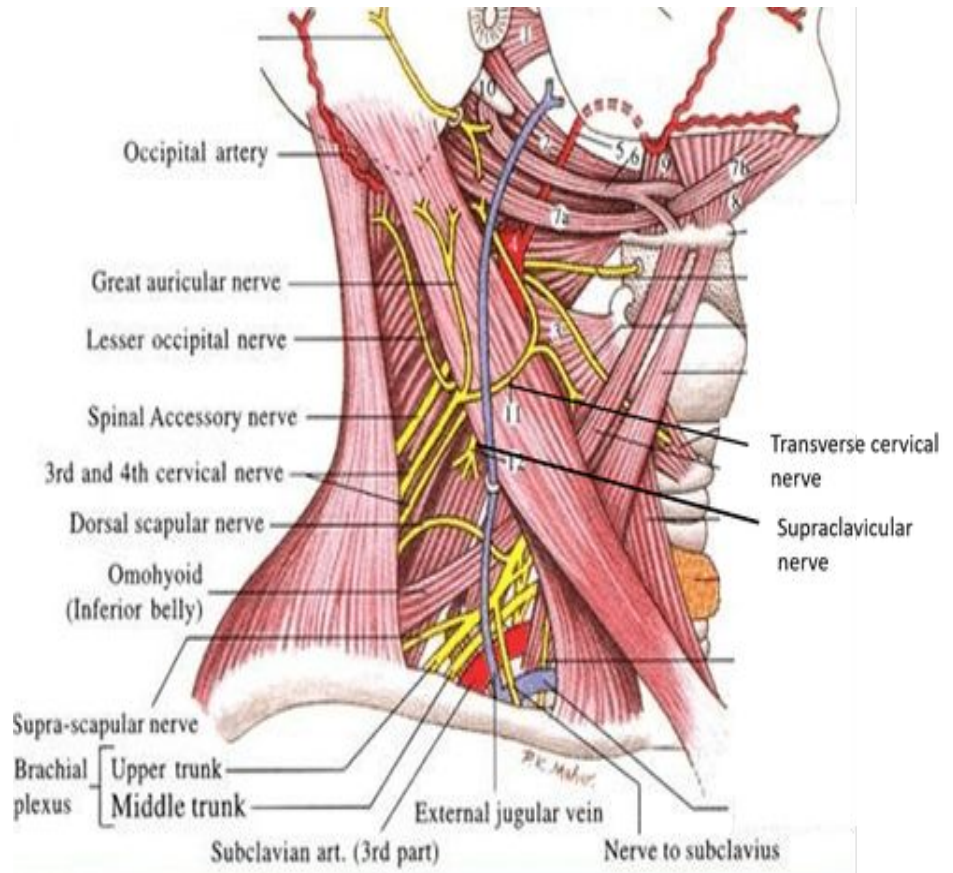
- Inferior belly of omohyoid divides the posterior triangle into:
  - a) A larger upper occipital triangle, &
  - b) A smaller lower supraclavicular or subclavian triangle



# Contents of the Occipital Triangle:

## A. Nerves:

- I. Spinal accessory nerve
- II. Cutaneous branches of cervical plexus:
  - i. Lesser occipital (C2)
  - ii. Great auricular (C2, C3)
  - iii. Anterior cutaneous nerve of neck (C2, C3)
  - iv. Supraclavicular nerves (C3, C4)
- III. Muscular branches:
  - i. Branches to levator scapulae (C3, C4)
  - ii. Branches to trapezius (C3, C4)
  - iii. Nerve to rhomboideus (C5)
- IV. Upper part of brachial plexus

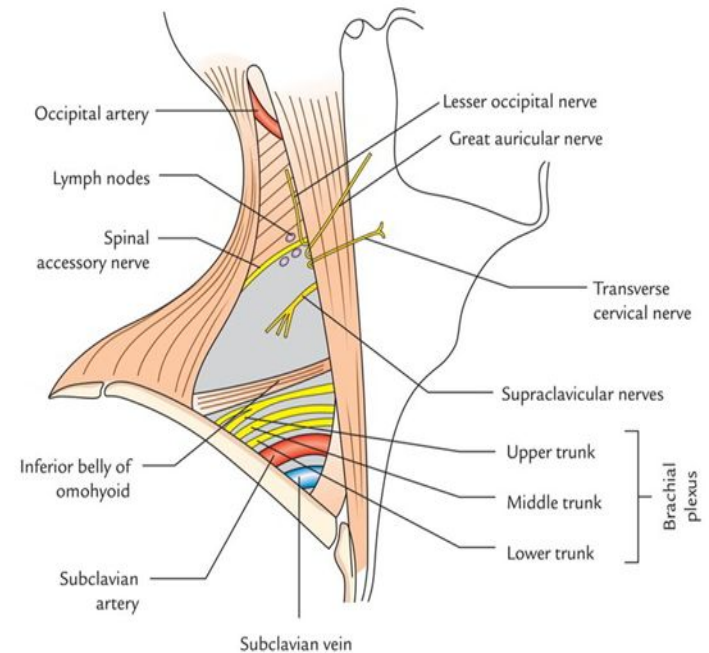


# Contents of the Occipital Triangle:

## B. Vessels:

- i. Transverse cervical artery and vein
- ii. Occipital artery

## C. Lymph nodes: along the posterior border of sternocleidomastoid, supraclavicular and occipital nodes

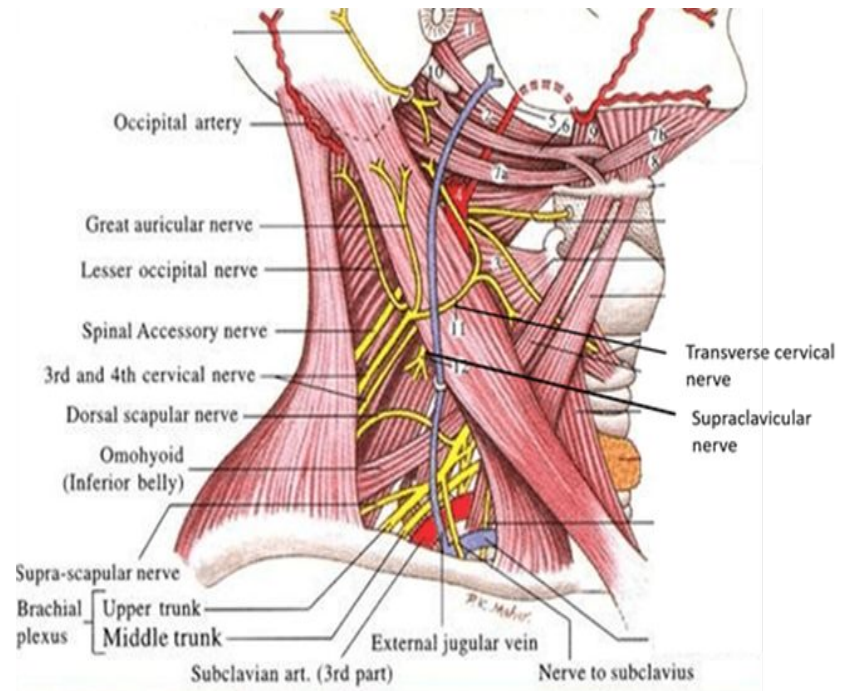




# Contents of the Subclavian Triangle:

## A. Nerves:

- I. Three trunks of brachial plexus
- II. Nerve to serratus anterior (long thoracic nerve-C5, C6, C7)
- III. Nerve to subclavius (C5, C6)
- IV. Suprascapular nerve (C5, C6)



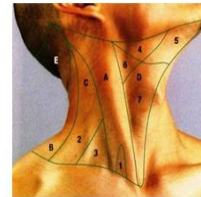
# Contents of the Subclavian Triangle:

## B. Vessels:

- i. Third part of subclavian artery and vein
- ii. Suprascapular artery and vein
- iii. Transverse cervical artery and vein
- iv. External jugular vein

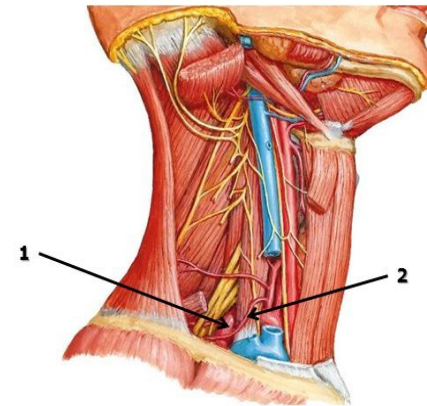
## C. Lymph nodes: few supraclavicular nodes

### CONTENTS OF THE SUBCLAVIAN TRIANGLE



#### OMOCLAVICULAR (SUBCLAVIAN) TRIANGLE

- 1. Subclavian artery (3<sup>rd</sup> part)
- 2. Suprascapular artery
- 3. Supraclavicular lymph nodes (not shown)



# Sternocleidomastoid Muscle

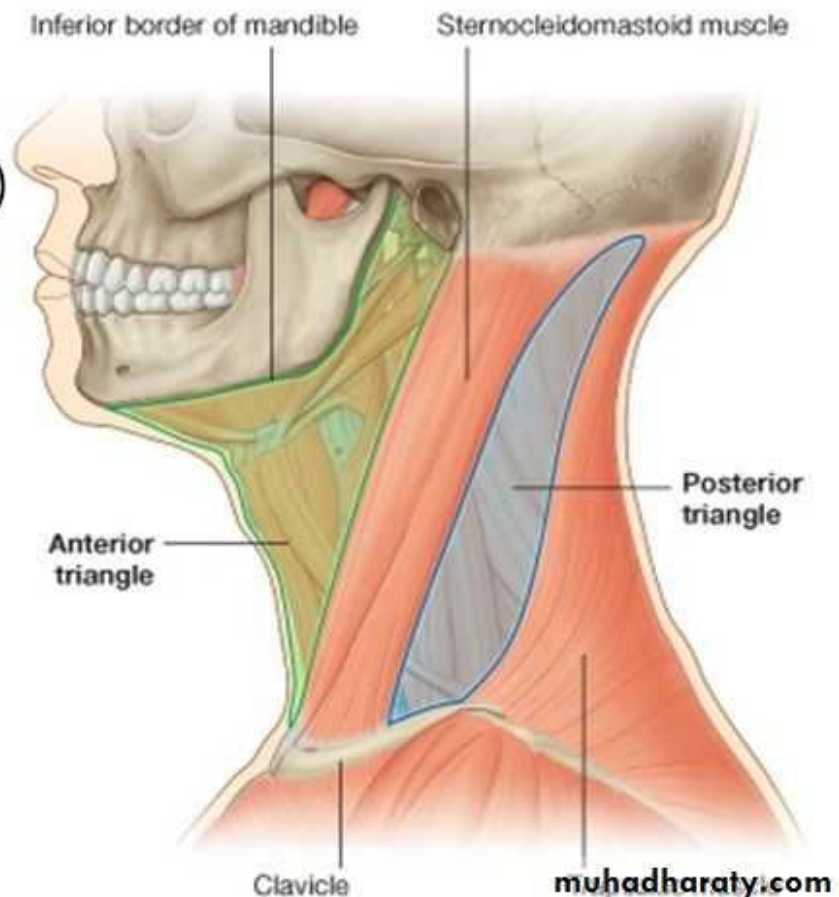
**It is the Key Muscle of the Neck** It divides the neck into anterior and posterior triangles. Its anterior border covers carotid arteries, internal jugular vein, and deep cervical lymph nodes; it also overlaps thyroid gland. It is covered superficially by skin, fascia, platysma muscle and external jugular vein. Deep surface of posterior border is related to cervical plexus of nerves, phrenic nerve, and upper part of brachial plexus.

**Origin:** Manubrium sterni (sternal head) and medial third of clavicle (clavicular head)

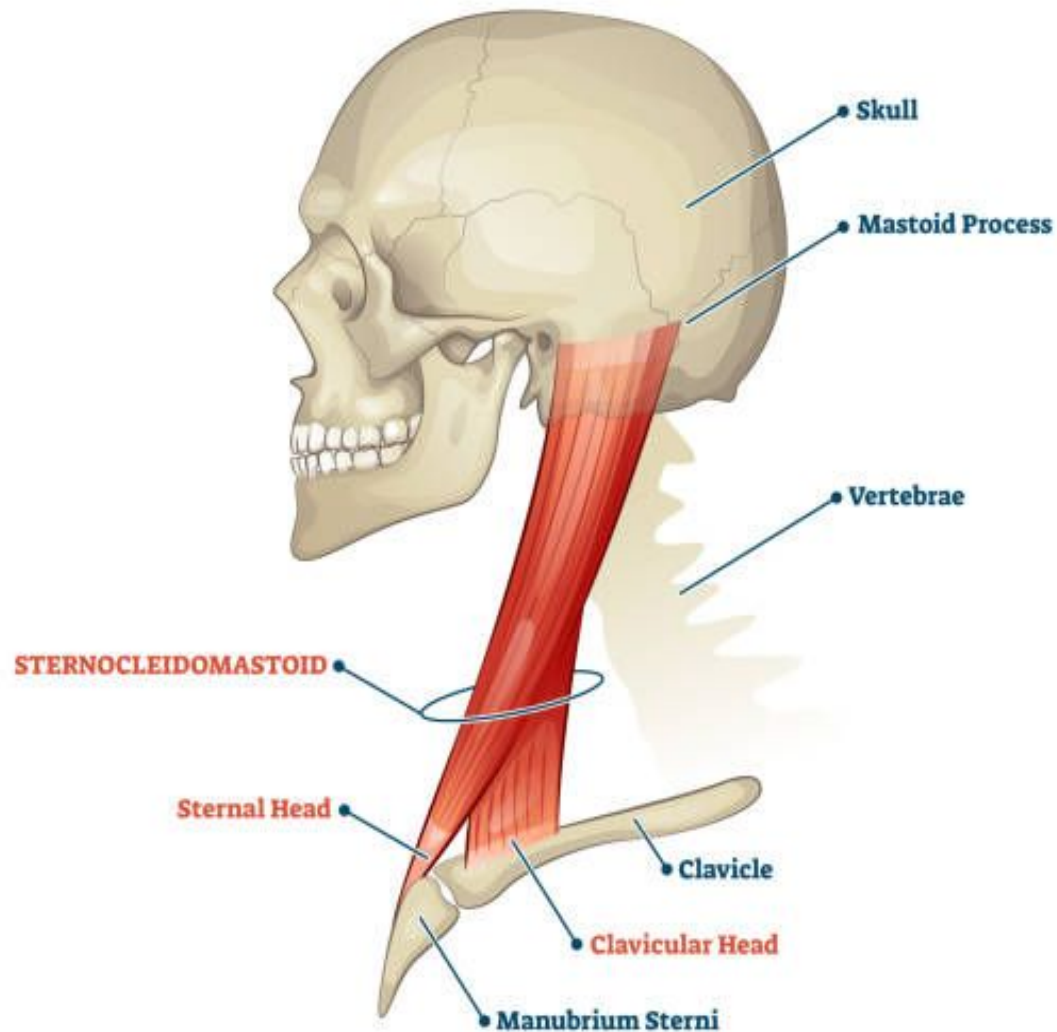
**Insertion:** Mastoid process of temporal bone and occipital bone

**Action:** Two muscles acting together flex neck draw head forward; one muscle rotates face to opposite side.

**Nerve Supply:** Spinal part of accessory nerve and anterior rami of C2 & C3.



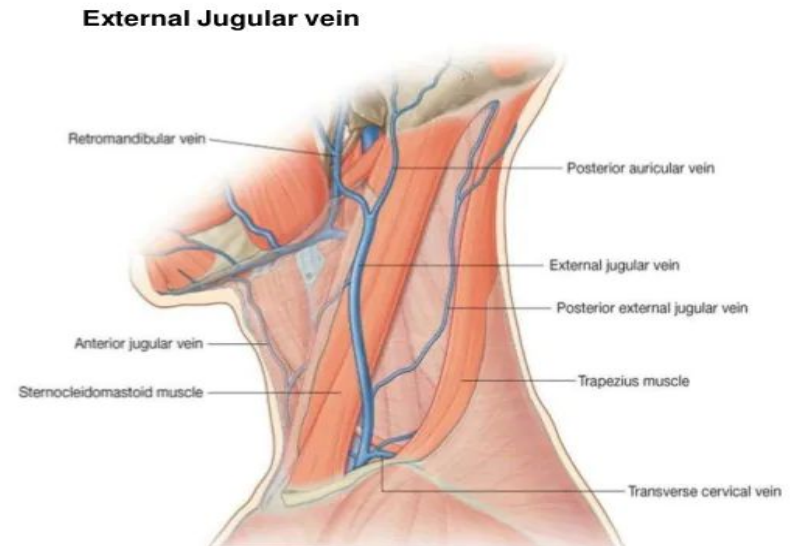
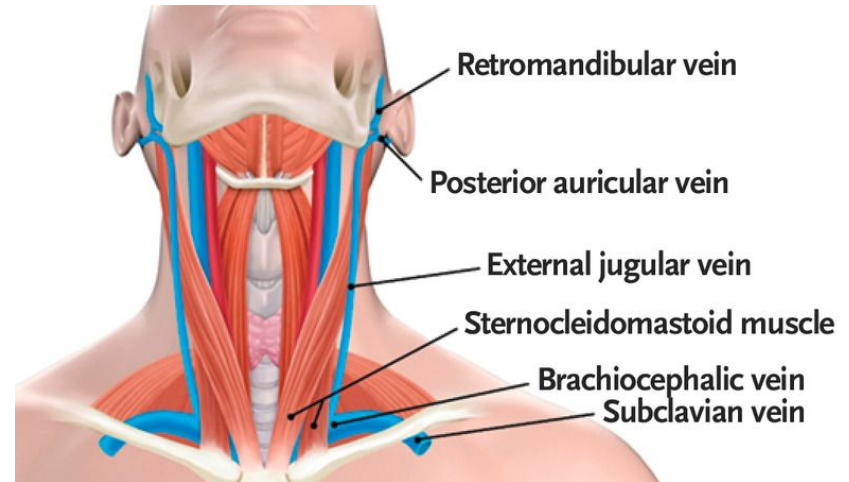
# STERNOCLEIDOMASTOID





# The External Jugular Vein:

- Lies deep to platysma
- Formed by the union of posterior auricular vein with posterior division of retromandibular vein
- It is valveless and used to assess jugular venous pressure
- **Tributaries:**
  - 1) Posterior external jugular vein
  - 2) Transverse cervical vein
  - 3) Suprascapular vein
  - 4) Anterior jugular vein





### Origin

- Occipital bone(external occipital protuberance),superior nuchal line, ligamentum nuchae, spine of seventh cervical vertebra, spines of all thoracic vertebrae and their supraspinous ligament

### Insertion

- Upper fibers into lateral third of clavicle; middle and lower fibers into acromion and spine of scapula

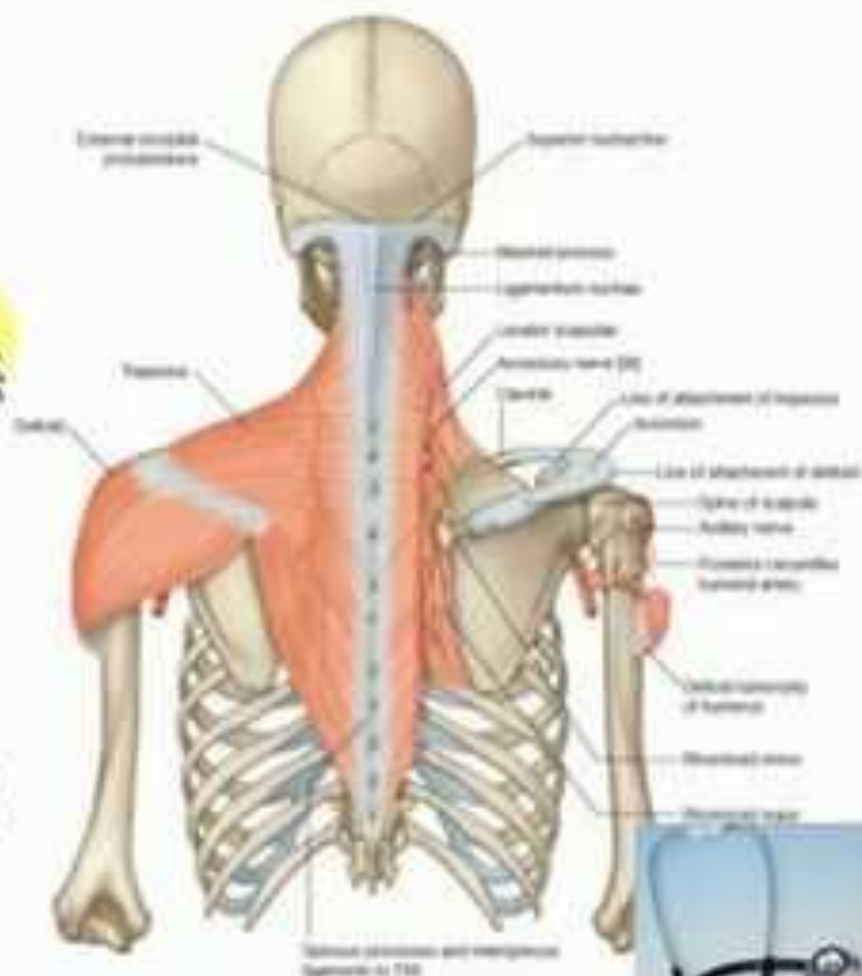
### Nerve Supply

- Spinal part of accessory nerve (motor) and C3 and 4 (sensory)
- **XI cranial nerve** (spinal part)

### Action

- Upper fibers elevate and upwardly rotate the scapula, extend the neck.
- middle fibers adduct(retracts) the scapula.
- lower fibers depress and help upper fibres in rotating scapula.

## Trapezius



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